**A**

**Project Phase-II Report**

Entitled

# Design and Performance Evaluation of an Intelligent Adaptive Digital Communication Receiver using Automatic Modulation Classification

*Submitted to the Department of Electronics Engineering in Partial Fulfilment for the Requirements for the Degree of*

**Bachelor of Technology**

**(Electronics and Communication Engineering)**

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**B. TECH. (EC), Final Year Project**

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(Year: 2026-27)

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# CERTIFICATE

This is to certify that the Project Phase-II Report entitled "**Design and Performance Evaluation of an Intelligent Adaptive Digital Communication Receiver using Automatic Modulation Classification**" is presented & submitted by Hari Gupta, Virav Vanani, Surendra Gamit, bearing Roll No. U23EC166, U23EC182, U23EC195 of B.Tech. Final Year in the partial fulfillment of the requirement for the award of B.Tech. Degree in Electronics & Communication Engineering for academic year 2026-27.

They have successfully and satisfactorily completed their **Project Phase-II Evaluation** in all respects. We certify that the work is comprehensive, complete and fit for evaluation.

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# Declaration

We hereby declare that the work presented in this report is original and has been carried out by us under the guidance of **Dr. Golak Santra**. The design, implementation, experiments, analysis, and documentation presented here are prepared for the Phase-II evaluation of the B.Tech. project. This report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

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# Abstract

Automatic modulation classification is useful in receivers that must operate without complete prior knowledge of the transmitted waveform. The objective of this Phase-II project is to extend modulation recognition into a complete adaptive digital communication receiver. Instead of ending at a modulation label, the proposed system uses the estimated format to configure a demodulator and evaluates whether the transmitted information is recovered correctly.

The transmitter generates random information bits and maps them to BPSK, QPSK, 8-PSK, or 16-QAM symbols. The symbols pass through a block-Rayleigh fading and additive-white-Gaussian-noise channel with carrier phase and frequency offsets. At the receiver, pilot-assisted synchronization and frame-level channel equalization are applied. A penalized nearest-constellation estimator selects the modulation, after which the corresponding demodulator recovers the bits. The experiment records bit error rate, symbol error rate, modulation-decision accuracy, packet success rate, and effective throughput over an SNR range of $-4$ to 16 dB.

The final experiment uses 200 frames per SNR and a fixed seed of 42. At 16 dB, the receiver achieves 0.095% BER, 0.172% SER, 100% modulation-decision accuracy, 75.5% packet success, and 1.888 bits/symbol effective throughput. The results show that an AMC decision must be evaluated through the complete information path because a correct format decision and error-free bit recovery are different requirements. The report documents the Phase-II theory, communication architecture, algorithms, experimental setup, measured results, limitations, and future SDR extensions.

# List of Abbreviations

| | |
|---|---|
| AMC | Automatic Modulation Classification |
| AWGN | Additive White Gaussian Noise |
| BER | Bit Error Rate |
| CSI | Channel State Information |
| I/Q | In-phase and Quadrature |
| PHY | Physical Layer |
| QAM | Quadrature Amplitude Modulation |
| PSK | Phase Shift Keying |
| RF | Radio Frequency |
| SDR | Software Defined Radio |
| SER | Symbol Error Rate |
| SNR | Signal-to-Noise Ratio |

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# Chapter 1
# Introduction

## 1.1 Motivation and Background

Modern wireless systems operate in congested and dynamic spectrum environments. The transmitter may change its modulation, the channel may introduce fading and phase rotation, and several users may share adjacent frequency bands. A conventional receiver is normally configured with the modulation and synchronization parameters before data reception begins. This assumption is unsuitable for a non-cooperative receiver, a cognitive-radio node, or a monitoring station that must process unknown signals.

Automatic modulation classification is the task of identifying the modulation format from received samples. It supports adaptive demodulation, spectrum monitoring, electronic support systems, interference analysis, and waveform-aware resource allocation. Classical approaches use likelihood tests, higher-order moments, cumulants, and cyclostationary features. These methods are physically interpretable, but their performance depends on the observation length, channel assumptions, and quality of synchronization [1, 2, 4].

Deep learning has provided another route by learning signal representations directly from I/Q samples. A neural classifier can identify local temporal patterns and phase relationships without requiring a separate handcrafted feature-design stage [7, 11]. However, a classification accuracy value alone does not demonstrate that a receiver can recover the transmitted information. The communication consequence of a wrong modulation decision must be measured after demodulation.

## 1.2 Applications of AMC

AMC is relevant to cognitive radio, where a receiver senses a waveform and selects a compatible processing chain. It is also useful in spectrum surveillance, where several unknown signals must be characterized from short observations. In adaptive wireless systems, modulation recognition can support mode selection, link adaptation, and interference-aware processing. In software-defined radio, the classifier can act as a control block between the digitized signal and a bank of candidate demodulators.

These applications impose requirements that are different from a static classification benchmark. The receiver must operate across channel conditions, report

confidence or uncertainty, and provide a measurable benefit to the recovered information. Phase II is designed around this system-level requirement.

## 1.3 Receiver Requirements

The receiver developed in this phase is required to perform more than a single classification operation. It must accept a stream of complex samples, estimate the modulation format, select a compatible demodulator, and report the quality of the recovered information. This requirement changes the way the experiment is designed. The transmitted bits must remain available as a reference, channel effects must be represented explicitly, and the output must include communication metrics rather than only a class label.

The receiver should also behave consistently when the signal quality changes. At high SNR, it should identify the modulation and recover most of the information bits. At low SNR, it should expose its uncertainty instead of producing a misleadingly confident output. For this reason, decision accuracy, BER, SER, packet success, and throughput are recorded separately. Together they show whether a recognition decision is useful to the physical layer.

## 1.4 Phase-I Baseline and Phase-II Extension

The earlier phase established a CNN-based AMC baseline using normalized I/Q windows from the RadioML dataset. That work answered the question of whether a compact model could distinguish several waveform classes. The baseline is retained in this report because it provides the starting point for the project and gives a reference for the learning components.

Phase II changes the endpoint of the system. The new transmitter produces known information frames, the channel introduces controlled impairments, and the receiver uses its modulation estimate to select a demodulator. The main new result is therefore not another isolated accuracy value. It is the measured behaviour of an adaptive link from transmitted bits to recovered bits. The two phases are related, but they answer different engineering questions and are reported separately.

## 1.5 Problem Definition and Objectives

**Problem Statement:** Design and evaluate an adaptive digital communication receiver that estimates the transmitted modulation and uses that estimate to recover information bits under noise, fading, phase offset, and frequency offset.

The objectives of the project are:

1. Generate reproducible information frames using BPSK, QPSK, 8-PSK, and 16-QAM.
2. Model a block-Rayleigh and AWGN channel with carrier impairments.
3. Apply pilot-assisted synchronization and channel equalization.
4. Estimate the modulation format using received constellation structure.
5. Select a compatible demodulator automatically.
6. Measure BER, SER, packet success, throughput, and decision accuracy.
7. Compare adaptive processing with a fixed-QPSK receiver.
8. Document the Phase-II implementation and experimental results clearly.

## 1.6 Contributions

The main contribution is a complete transmitter–channel–receiver simulation in which modulation recognition controls demodulation. The work adds four digital modulation formats, configurable channel impairments, frame-level correction, an interpretable estimator, adaptive bit recovery, and a fixed-modulation reference. It also introduces communication-level evaluation rather than reporting only a class label.

## 1.7 Scope and Limitations

The current work is a controlled software simulation. It includes a flat block-fading model, AWGN, carrier phase and frequency offsets, and pilot-assisted correction. Pulse shaping, matched filtering, timing-loop implementation, forward-error correction, interleaving, retransmission, and over-the-air SDR measurements are outside the present scope. These limitations are stated so that the simulation results are not confused with hardware measurements.

## 1.8 Organization of the Report

Chapter 2 reviews classical AMC, deep-learning approaches, communication receiver design, and the research gap. Chapter 3 presents the mathematical background of complex baseband signals, digital modulation, channel impairments, synchronization, and evaluation metrics. Chapter 4 describes the generated Phase-II data, modulation classes, preprocessing, and experimental record. Chapter 5

explains the proposed receiver methodology and design. Chapter 6 describes implementation and experimental setup. Chapter 7 presents measured results and discussion. Chapter 8 concludes the work and identifies future extensions.

## 1.9 Chapter Summary

Chapter 1 establishes the motivation, project objectives, and distinction between the two project phases. Chapter 2 places the work in the context of classical AMC, learning-based recognition, and adaptive receiver design. Chapter 3 develops the signal, channel, synchronization, and metric equations. Chapter 4 describes how the communication frames are generated and prepared. Chapter 5 gives the proposed receiver in block form. Chapter 6 records the software and experiment settings. Chapter 7 presents the numerical results, including the complete SNR sweep and the retained Phase-I comparison. Chapter 8 discusses the contribution, limitations, and next implementation steps.

# Chapter 2
# Literature Review

## 2.1 Classical AMC Approaches

Early AMC systems used statistical moments, higher-order cumulants, likelihood functions, and cyclostationary features. Higher-order statistics can distinguish modulation families because the distribution of a received signal changes with the waveform. Cyclostationary methods exploit periodic features introduced by symbol timing, carrier structure, or framing [4, 5].

The advantage of classical methods is interpretability. A designer can identify which feature is responsible for a decision and can often estimate the computational complexity analytically. Their limitation is sensitivity to observation length and model mismatch. Noise, fading, carrier offset, frequency-selective propagation, and nonlinear hardware can alter the statistics on which the decision depends.

## 2.2 Likelihood and Feature-Based Methods

Likelihood-based AMC compares the received sequence with a probability model for every candidate modulation. When the channel and noise parameters are known, such methods can be statistically efficient. In practical systems, the required likelihood model is difficult to maintain because the receiver may not know the exact fading, synchronization, or hardware condition.

Feature-based systems reduce the computational burden by extracting moments, spectral features, cyclic features, or instantaneous amplitude and phase statistics. They can provide a useful compromise between complexity and performance, but their success depends on choosing features that remain discriminative at low SNR. Phase II uses a constellation-based estimator because it is transparent and directly connected to demodulator selection.

## 2.3 Deep Learning for AMC

Deep learning changed the AMC workflow by allowing features to be learned from raw I/Q sequences. Convolutional neural networks are attractive because one-dimensional filters can detect local waveform patterns, short phase transitions, and amplitude relationships. RadioML-style datasets have provided a common basis for studying these models [7, 8].

The main limitation of a classifier-only evaluation is that the output is often treated as the final result. A high classification accuracy does not directly specify the BER after a receiver uses the decision. Conversely, an incorrect decision may select an incompatible demodulator and cause a much larger information loss than a single classification error suggests.

## 2.4 Adaptive Receiver Design

An adaptive receiver contains a sensing or recognition stage followed by a configuration stage. The configuration stage may select a demodulator, symbol mapping, coding mode, timing method, or channel-estimation strategy. This makes receiver performance dependent on both recognition and subsequent signal processing.

The Phase-II design follows this principle. The estimator does not operate as an isolated classifier. Its output selects one of four demodulators, and the receiver compares the recovered bits with the transmitted bits. This creates a traceable relationship between waveform uncertainty and communication reliability.

## 2.5 Over-the-Air Generalization

Synthetic experiments provide repeatability and controlled comparison, but they do not include every effect present in an RF front-end. Oscillator phase noise, automatic-gain-control transients, antenna response, analog filtering, frequency-selective multipath, and timing drift can reduce performance when a system moves to an SDR. Over-the-air studies have shown the importance of these effects [8].

The present report therefore treats the simulation as an engineering validation stage. The channel model is sufficiently rich to expose modulation-decision and bit-recovery failures, while remaining reproducible for a project evaluation. Hardware validation is proposed as the next stage.

## 2.6 Benchmark Datasets and Tooling

RadioML 2016.10a is a widely used benchmark for raw-I/Q modulation classification. It is useful for retaining an earlier AMC baseline, but the new Phase-II link is evaluated with generated information frames so that BER and packet success can be measured directly. Python, NumPy, Matplotlib, and PyTorch provide the numerical and visualization environment used by the project [14].

## 2.7 Summary of Representative Studies

Table 2.1: Representative studies and their relation to the Phase-II work.

| **Reference** | **Approach** | **Relevance to this project** |
|---|---|---|
| Gardner [5] | Cyclostationary analysis | Provides interpretable signal features for detection. |
| Azzouz and Nandi [4] | Statistical AMC | Shows the importance of feature and channel assumptions. |
| O'Shea and Corgan [7] | CNN on raw I/Q | Motivates data-driven waveform recognition. |
| O'Shea *et al.* [8] | Over-the-air classification | Demonstrates the need for realistic validation. |
| This project | Adaptive communication receiver | Connects modulation estimation to demodulation and BER. |

## 2.8 Research Gap

Many AMC studies end after predicting a modulation label, while many link simulations assume that the modulation is already known. The engineering gap between these two stages is whether the recognition result can configure a useful receiver and deliver reliable information. Phase II addresses this gap by evaluating the complete path from generated bits to recovered bits.

## 2.9 Comparison of Evaluation Practices

Three evaluation practices can be distinguished in the reported literature. The first reports only a classifier accuracy, which is useful for comparing recognition models but does not show how the decision affects a receiver. The second assumes perfect modulation knowledge and evaluates BER, which gives a clear link result but removes the recognition problem. The third combines recognition with demodulation and evaluates the recovered information. The present project follows the third practice.

This distinction is important when comparing numerical values. A classifier accuracy near one does not imply a BER near zero because symbol decisions still depend on residual noise, gain error, phase error, and the separation between

constellation points. Similarly, a low BER averaged over all symbols may still produce a low packet-success rate when a long frame is declared unsuccessful after one error. The report therefore avoids using one metric as a substitute for the complete receiver result.

## 2.10 Design Choices from the Review

The review led to four design choices. First, the modulation set contains both phase-only formats and a rectangular amplitude-and-phase format. This allows the receiver to be tested on different decision geometries. Second, the channel contains both additive noise and multiplicative fading so that the estimator is not tested only on an ideal amplitude scale. Third, a fixed-QPSK path is included as a simple reference rather than as a competing learning model. Finally, the estimator is based on constellation distance, which makes each decision traceable and leaves a clear path for replacing it with a trained model in future work.

## 2.11 Summary

The literature supports the need for automatic modulation recognition, but it also shows that a classification result is only one part of a receiver. The Phase-II work retains the useful recognition perspective from Phase I and adds a controlled communication link around it. This arrangement makes the effect of modulation estimation visible in the final bit and packet metrics.

# Chapter 3
# Theoretical Background

## 3.1 Baseband Signal Model

A passband signal can be represented in complex baseband form as

$$x[n] = I[n] + jQ[n], \tag{3.1}$$

where $I[n]$ and $Q[n]$ are the in-phase and quadrature components. The complex representation preserves amplitude and phase information while allowing the transmitter, channel, and receiver to be implemented as discrete-time operations.

For a frame of $N$ symbols, the transmitter generates $x[0], x[1], \ldots, x[N-1]$. The receiver observes a corrupted sequence and must estimate the format before applying a symbol-to-bit mapping. The quality of this estimate affects every subsequent stage.

## 3.2 Digital Modulation Families

BPSK uses two opposite phase points and carries one bit per symbol. Its large decision separation makes it robust, but its spectral efficiency is low. QPSK uses four phase points and carries two bits per symbol, providing a common balance between robustness and rate. 8-PSK carries three bits per symbol by reducing angular separation. 16-QAM uses both amplitude and phase and carries four bits per symbol, but requires more accurate gain, phase, and noise control.

Table 3.1: Digital modulation formats used by the Phase-II receiver.

| **Format** | **Order** | **Bits/symbol** | **Characteristic** |
|---|---|---|---|
| BPSK | 2 | 1 | Two phase states and large decision margin. |
| QPSK | 4 | 2 | Four phase states and moderate spectral efficiency. |
| 8-PSK | 8 | 3 | Higher rate with reduced angular separation. |
| 16-QAM | 16 | 4 | Amplitude and phase states for high rate. |

## 3.3 Channel Impairments

The Phase-II received signal is modelled as

$$y[n] = h\, x[n] e^{j(2\pi \Delta f n + \phi)} + w[n], \tag{3.2}$$

where $h$ is a complex block-Rayleigh channel coefficient, $\Delta f$ is a normalized frequency offset, $\phi$ is a carrier phase offset, and $w[n]$ is complex AWGN. The coefficient is constant during one frame and changes between frames, allowing fading to affect packet-level performance.

Rayleigh fading changes the amplitude and phase of the signal without a dominant line-of-sight component. AWGN introduces random perturbations whose variance is controlled by the selected SNR. Phase offset rotates the entire constellation, while frequency offset produces a rotation that changes with sample index. These impairments test different aspects of receiver correction and constellation recognition.

## 3.4 Signal-to-Noise Ratio

The signal-to-noise ratio is defined as

$$\mathrm{SNR}_{dB} = 10 \log_{10} \left( \frac{P_s}{P_n} \right), \tag{3.3}$$

where $P_s$ is the average signal power and $P_n$ is the average noise power. The simulation normalizes the candidate constellations and scales complex Gaussian noise from the selected SNR so that the modulation formats are compared under one common convention.

## 3.5 Synchronization and Equalization

In a practical receiver, synchronization includes automatic gain control, timing recovery, carrier-frequency estimation, carrier-phase recovery, and channel estimation. Phase II uses a pilot-assisted simulation estimate for the frame-level carrier and channel parameters. This isolates the new adaptive-modulation problem from an unfinished hardware synchronization loop.

After carrier correction, the frame is equalized using

$$\hat{x}[n] = \frac{y[n]}{h}. \tag{3.4}$$

The corrected samples are normalized before their distance to each candidate constellation is calculated. The limitation of this controlled estimate is explicitly discussed in the results chapter.

## 3.6 AMC Approaches

The project distinguishes three receiver actions. A fixed receiver applies one known demodulator to every frame. A classifier-only receiver produces a label but does not recover information. The proposed adaptive receiver estimates the format and passes the decision to the corresponding demodulator. This third arrangement is the focus of Phase II.

## 3.7 Communication Metrics

The bit error rate and symbol error rate are

$$\mathrm{BER} = \frac{N_{\mathrm{bit,error}}}{N_{\mathrm{bit,total}}}, \qquad \mathrm{SER} = \frac{N_{\mathrm{symbol,error}}}{N_{\mathrm{symbol,total}}}. \tag{3.5}$$

A packet is successful only when every bit in its frame is recovered correctly. If $b_m$ is the bits-per-symbol value of the selected modulation, the effective throughput is the average of $b_m$ multiplied by the packet-success indicator. Therefore, packet success is stricter than BER and exposes the practical effect of even a small number of bit errors.

## 3.8 Bit Mapping and Symbol Decisions

For a modulation order $M$, the input stream is divided into groups of $\log_2 M$ bits. Each group is assigned to one point in the normalized constellation. The same mapping is used in reverse at the receiver after a nearest-point decision. Using one mapping definition in both directions avoids an artificial error caused by a mismatch between the transmitter and demodulator.

The minimum distance between constellation points is a useful qualitative indicator of robustness. BPSK has the largest separation for its rate, while 8-PSK packs more points around the same phase circle. 16-QAM provides four bits per symbol, but its inner points can be confused with adjacent points after noise and amplitude distortion. These differences explain why one common SNR does not produce the same BER for all formats.

## 3.9 Frame-Level Reliability

If the probability of a bit error is $p_b$ and a frame contains $L$ bits, the idealized probability of a completely correct frame is approximately

$$P_{\text{packet}} \approx (1 - p_b)^L. \tag{3.6}$$

The approximation assumes independent bit errors and is not used as a replacement for the measured packet count. It explains, however, why a BER of a few percent can coexist with a packet-success rate near zero for a long frame. In the experiment, the frame length changes with modulation order because the symbol count is fixed, so the measured packet metric also reflects the information load of each format.

## 3.10 Throughput Definition

The nominal information rate of a modulation is its bits-per-symbol value. The effective value reported in this project multiplies that value by the packet-success indicator before averaging over frames:

$$T_{\text{eff}} = \frac{1}{F} \sum_{i=1}^{F} b_i \mathbb{1}\{\text{frame } i \text{ is correct}\}. \tag{3.7}$$

This definition does not include coding overhead, pilot overhead, filtering loss, or hardware latency. It is intended to compare the useful information delivered by the adaptive and fixed receivers under the same simulation conditions.

## 3.11 Interpretation of Decision Accuracy

Decision accuracy measures whether the estimated modulation label matches the transmitted label. It does not measure whether all symbols are correct. For example, at 10 dB the measured decision accuracy is 99.5 percent, while the BER is 4.69 percent. This difference is expected because the estimator can identify the family correctly while individual noisy samples cross decision boundaries. The separation of these two measurements is one of the main reasons for using a complete link experiment.

# Chapter 4
# Dataset and Preprocessing

## 4.1 Phase-II Data Generation

Unlike a classification-only experiment, the Phase-II link begins with information bits rather than pre-labelled signal windows. For every frame, a reproducible random generator creates a binary sequence. One of the four candidate modulation formats is selected, and the bits are grouped according to its order. The resulting complex symbols are retained as the transmitter reference for later error calculation.

Each frame contains 256 symbols. Thus, BPSK frames contain 256 information bits, QPSK frames contain 512 bits, 8-PSK frames contain 768 bits, and 16-QAM frames contain 1024 bits. Using the same symbol-frame length makes packet-level comparisons consistent while allowing the information rate to change with modulation order.

## 4.2 Modulation Classes

Table 4.1: Modulation classes used to generate Phase-II communication frames.

| **Class** | **Bits/symbol** | **Use in the experiment** |
|---|---|---|
| BPSK | 1 | Robust low-order reference. |
| QPSK | 2 | Fixed-receiver comparison format. |
| 8-PSK | 3 | Higher-order phase modulation. |
| 16-QAM | 4 | High spectral-efficiency format. |

## 4.3 Constellation Normalization

The constellation points are normalized to a common average-power convention. This avoids giving one format an artificial advantage because of a larger absolute amplitude. After channel correction, the received symbols are placed on the same scale before nearest-point distances are compared.

## 4.4 Channel Data Preparation

For each generated frame, the channel draws a block-Rayleigh coefficient, carrier phase, frequency offset, and complex AWGN realization. The SNR is selected from the sweep

$$-4, -2, 0, 2, 4, 6, 8, 10, 12, 14, 16 \text{ dB}.$$

The transmitted bits, transmitted symbols, received samples, estimated format, recovered bits, and error counts are retained until all frame-level metrics have been calculated.

## 4.5 Preprocessing at the Receiver

The receiver applies the inverse simulated carrier rotation and equalizes the block channel before estimation. The corrected samples are normalized using the candidate constellation scale. This operation is not a replacement for a practical timing loop; it is a controlled preprocessing stage that makes the contribution of adaptive modulation measurable.

## 4.6 Experiment Split and Repetition

The communication experiment is not divided into train, validation, and test subsets because it is a link simulation rather than a supervised training run. Instead, every SNR value is evaluated using 200 independently generated frames with seed 42. The same modulation list, frame length, channel model, and metric definitions are used at every SNR.

## 4.7 Data Integrity and Reproducibility

The complete numerical arrays are written to a JSON record. This record contains the SNR values, adaptive and fixed-receiver BER and SER, decision accuracy, packet success, throughput, modulation list, frame count, and seed. Figures are generated from the same record, preventing manual transcription errors between the experiment and the report.

## 4.8 Limitations of the Generated Data

The data is generated by a controlled software channel and therefore does not include antenna response, analog filtering, oscillator phase noise, frequency-selective

multipath, timing drift, or hardware nonlinearity. The generated frames are appropriate for algorithm verification and comparative analysis, but a future SDR experiment is required before making claims about over-the-air performance.

## 4.9 Data Record Structure

Every frame has a corresponding record containing the selected modulation, SNR, channel realization, transmitted bits, received samples, estimated modulation, recovered bits, and error counts. The experiment driver reduces these frame records into arrays indexed by SNR. Keeping the frame-level information during the experiment is useful for diagnosing unusual results, such as a high decision accuracy together with a low packet-success rate.

The final JSON file stores the aggregate values used in the report. It also stores the modulation names and the experiment seed, so the order of the curves is not dependent on the order in which a plotting script happens to read a dictionary. The PNG figures are presentation views of this record; they are not manually redrawn result diagrams.

## 4.10 Normalization Checks

Before the channel is applied, the average power of each normalized constellation is checked. This check is important because the SNR sweep is meaningful only when the formats are compared at a common reference power. After equalization, the received samples are normalized using the same convention before distance scores are calculated. A gain mismatch would otherwise make the estimator prefer a format for amplitude reasons rather than for its geometric structure.

## 4.11 Quality-Control Cases

Three simple cases are used before the full sweep. In the first case, a clean frame is modulated and demodulated without a channel; all recovered bits must match. In the second case, an invalid modulation name or an incompatible bit length must produce an explicit validation error. In the third case, a clean constellation is supplied to the estimator; the expected class must be selected. These cases do not replace the noisy experiment, but they separate implementation errors from channel-induced errors.

# Chapter 5
# Proposed Methodology and Model Design

## 5.1 End-to-End Pipeline

The proposed Phase-II pipeline is

$$\begin{aligned}\text{random bits} \rightarrow \text{modulator} \rightarrow \text{channel} \rightarrow \text{correction} \rightarrow \text{AMC decision} \rightarrow \\ \text{adaptive demodulator} \rightarrow \text{recovered bits.}\end{aligned} \tag{5.1}$$

The final bit sequence is compared with the transmitter reference. This pipeline is intentionally different from a classifier-only workflow because the modulation decision controls a subsequent physical-layer operation.

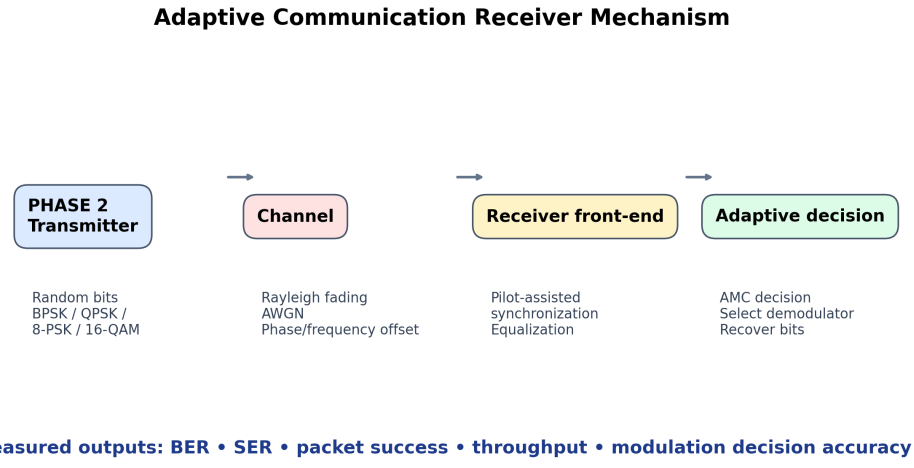



Figure 5.1: End-to-end Phase-II adaptive communication receiver.

## 5.2 Transmitter Model

The transmitter selects a modulation and groups the random bits into symbols. The mapper produces normalized complex points according to the selected constellation. The mapping is deterministic and is reused by the corresponding demodulator, so a clean round-trip test should recover all bits exactly.

## 5.3 Channel Model

The channel applies block fading, carrier rotation, and AWGN in sequence. Block fading changes the amplitude and phase of a complete frame. Carrier phase offset rotates the constellation, while frequency offset causes a progressive rotation. AWGN controls the SNR-dependent uncertainty that eventually produces both estimator errors and symbol errors.

## 5.4 Receiver Front-End

The receiver front-end reverses the simulated carrier rotation and divides by the block channel coefficient. The output is a corrected complex sequence. In an SDR implementation, this stage would be replaced by a timing recovery loop, pilot detector, carrier-recovery loop, and channel estimator. The current design keeps the estimate controlled so that the adaptive demodulation contribution can be isolated.

## 5.5 Constellation-Distance Estimator

Let $\mathcal{C}_m$ denote the normalized constellation for candidate format $m$. For each corrected received sample, the estimator finds the nearest point in $\mathcal{C}_m$ and calculates the average distance $d_m$. It then selects

$$\hat{m} = \arg\min_{m} \left( d_m + \lambda \log_2 |\mathcal{C}_m| \right), \tag{5.2}$$

where the penalty discourages a high-order constellation from winning only because it contains more possible points.

## 5.6 Adaptive Demodulator

The estimated format selects one of four demodulation rules. Each corrected sample is assigned to its nearest point, and the point index is converted back to bits using the transmitter's mapping. The recovered bits are aligned with the original frame and compared bit by bit. The estimator decision is recorded separately so that decision errors and within-format detection errors can be distinguished.

## 5.7 Fixed-QPSK Baseline

The fixed-QPSK receiver applies the QPSK demodulator regardless of the transmitted format. It represents a system that does not perform modulation adaptation. Its performance is expected to remain poor for BPSK, 8-PSK, and 16-QAM frames, but it provides a useful reference for measuring the value of adaptive selection.

## 5.8 Failure Propagation

The system has a clear failure chain. At low SNR, constellation points overlap and the estimator may choose the wrong format. The wrong format selects an incompatible demodulator, which can create many bit errors. Even when the estimated format is correct, noise can move individual points across decision boundaries. A single bit error is sufficient to mark a complete packet unsuccessful.

## 5.9 Design Rationale

The Phase-II estimator is deliberately interpretable. The candidate constellations, distance scores, selected modulation, recovered symbols, and final bit metrics can all be inspected. This design also provides a baseline for future replacement by a trained CNN or a confidence-aware estimator.

## 5.10 Detailed Receiver Sequence

The receiver processes one frame in a fixed order. It first removes the known simulated carrier rotation and compensates the block channel. It then computes candidate distances for all four constellations. The minimum penalized distance determines the modulation estimate. Only after this decision is made does the receiver select the corresponding bit mapper. Finally, the recovered bit sequence is compared with the transmitter reference and the frame-level counters are updated.

This sequence prevents the experiment from accidentally using the true transmitted modulation during demodulation. The true label is used only for measuring decision accuracy and for calculating the reference errors. The demodulator receives the estimate produced by the receiver, which makes the adaptive path a genuine decision-dependent pipeline.

## 5.11 Estimator Penalty and Bias

The number of points in a constellation affects the average nearest-point distance. A high-order format has more candidate points and can sometimes obtain a lower unpenalized distance even when the received samples are better explained by a lower-order format. The penalty term in the estimator reduces this bias by adding a cost proportional to the logarithm of the constellation order.

The penalty is not intended to be a universal statistical criterion. It is a simple engineering correction that keeps the estimator interpretable. Its value is held constant over the SNR sweep so that the improvement at high SNR is not produced by changing a tuning parameter at each point. A future version can select this parameter using a separate validation experiment or replace the distance rule with a learned confidence model.

## 5.12 Error Propagation Paths

There are two main paths from a noisy sample to an incorrect frame. In the first path, the constellation estimator selects the wrong format. The demodulator then uses an incompatible geometry and a large number of bits can be wrong even when individual samples are not extremely noisy. In the second path, the format estimate is correct but one or more samples cross a decision boundary. This produces symbol and bit errors without reducing decision accuracy.

## 5.13 Complexity Considerations

For a frame containing $N$ received symbols and candidate constellations $\mathcal{C}_m$, the distance calculation requires approximately

$$\mathcal{O}\left(N \sum_m |\mathcal{C}_m|\right)$$

complex comparisons. With the four formats used here, the largest term is the 16-QAM candidate set. The implementation is therefore suitable for offline experiments and small real-time demonstrations. A deployed receiver could reduce the cost using a hierarchical classifier, moment-based pre-screening, or a neural model followed by a small verification step.

# Chapter 6
# Implementation and Experimental Setup

## 6.1 Software and Tools

The project is implemented in Python using NumPy for numerical operations, Matplotlib for figures, PyTorch for the retained learning components, and pytest for automated verification. The source is organized into reusable communication primitives, an experiment driver, visualization utilities, tests, and report artifacts. The report does not include executable command blocks; the repository documentation contains the operational instructions.

## 6.2 Experimental Environment

The final communication experiment is deterministic at the experiment level. The random seed is 42, the frame length is 256 symbols, and 200 frames are processed at each SNR. The channel and estimator parameters are held constant across the sweep, allowing changes in the curves to be interpreted primarily as SNR effects.

## 6.3 Hyperparameters and Link Parameters

## 6.4 Evaluation Metrics

Five primary metrics are calculated. BER counts erroneous bits, SER counts erroneous symbols, and modulation-decision accuracy counts correct format estimates. Packet success requires every bit in a frame to be correct. Effective throughput weights the bits-per-symbol value of each successful frame by its packet-success indicator. These metrics are reported together because no single value captures the complete receiver behavior.

## 6.5 Experimental Procedure

At each SNR, the experiment generates 200 frames. For every frame it selects a modulation, creates bits, maps them to symbols, applies the channel, corrects the received sequence, estimates the modulation, demodulates the symbols, and calculates errors. The same transmitted frame is also processed by the fixed-QPSK reference. Aggregate arrays are saved before figures are created.

Table 6.1: Phase-II link parameters used for the final experiment.

| **Parameter** | **Value** |
|---|---|
| Modulation formats | BPSK, QPSK, 8-PSK, 16-QAM |
| Symbols per frame | 256 |
| SNR range | $-4$ to 16 dB |
| SNR increment | 2 dB |
| Frames per SNR | 200 |
| Random seed | 42 |
| Channel | Block Rayleigh fading and AWGN |
| Carrier impairments | Phase and frequency offset |
| Correction | Pilot-assisted simulation estimate |
| Estimator | Penalized nearest-constellation distance |

## 6.6 Verification

Automated tests cover clean modulation round trips, validation of invalid inputs, deterministic data handling, and clean constellation identification. The report source is checked for balanced LaTeX environments, existing figure references, and the required style dependencies. These checks supplement the numerical experiment.

## 6.7 Reproducibility Record

The JSON result record stores the seed, frame count, SNR list, modulation list, adaptive metrics, and fixed-QPSK metrics. The figures are generated from the same values used in the tables.

## 6.8 Implementation Modules

The implementation is divided into modulation, channel, receiver, metric, and experiment-driver responsibilities. The modulation functions define the constellations and bit mappings. The channel functions apply fading, carrier offset, and noise. The receiver functions perform correction, estimate the format, and demodulate. The experiment driver repeats these operations over the SNR list and

writes the aggregate record.

This separation is useful during debugging because each stage can be tested without running the complete sweep. It also makes the Phase-II design different from a single notebook-style experiment. The report presents the algorithmic behaviour without reproducing source-code listings, while the repository retains the implementation files used to produce the numerical results.

## 6.9 Numerical Conventions

Complex samples use NumPy complex arrays and the real and imaginary parts represent the in-phase and quadrature components. Random numbers are generated from a fixed seed at the experiment level. SNR values are stored in decibels, while noise scales are calculated in linear power units before being applied to the complex samples. Percentages in the report are rounded for readability; the JSON record retains the underlying decimal values.

## 6.10 Parameter Selection

The frame length of 256 symbols gives enough samples for a constellation estimate while keeping one experiment run manageable. Two hundred frames per SNR provide a meaningful packet-success count without making the experiment dependent on a single channel realization. The SNR interval from $-4$ to 16 dB covers the transition from an unusable low-SNR link to a high-SNR link where modulation decisions are stable.

## 6.11 Setup Limitations

Repeating the experiment with the same seed reproduces the stored numerical record. Repeating it with a different seed is expected to change individual frames and slightly change the aggregate values, especially packet success, because packet success is a binary frame-level metric. The experiment is therefore deterministic for comparison, but the result should not be read as an exact performance guarantee for every random channel realization.

## 6.12 Runtime and Resource Considerations

The communication experiment is dominated by the repeated distance calculation for the candidate constellations. Its memory requirement is small because

one frame is processed before the aggregate arrays are updated. This makes the experiment suitable for an ordinary project computer and avoids making the reported result dependent on a specialised GPU. The retained CNN baseline and the Phase-II communication link are therefore kept as separate experiments with separate resource requirements.

The same implementation can be extended in two directions. More frames can be evaluated to improve statistical resolution, or more candidate formats and channel models can be added to test generality. In both cases the result record should be extended before the report tables are updated so that every new curve remains traceable to an experiment setting.

# Chapter 7
# Results and Discussion

## 7.1 Training Dynamics

Phase II is evaluated as a communication-link experiment rather than as a new supervised training run. Consequently, the primary result is the SNR-dependent receiver behavior, not a new training-loss curve. The retained recognition baseline motivates the AMC function, while the new evidence is the measured relationship between modulation selection, bit recovery, packet success, and throughput.

## 7.2 Retained Phase-I Learning Results

The Phase-I CNN baseline was trained on normalized I/Q windows from the RadioML 2016.10a dataset. It achieved 53.94 percent accuracy on the held-out test split, with the expected increase in accuracy as the SNR improved. The baseline remains relevant because it demonstrates the recognition component that motivated the adaptive receiver, but it is not used to claim the communication-link results reported below.

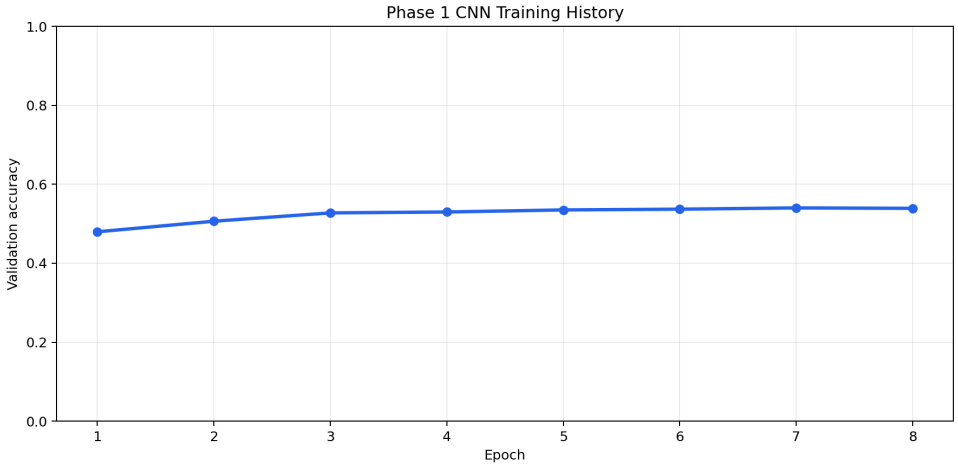



Figure 7.1:

Training and validation history retained from the Phase-I CNN baseline.

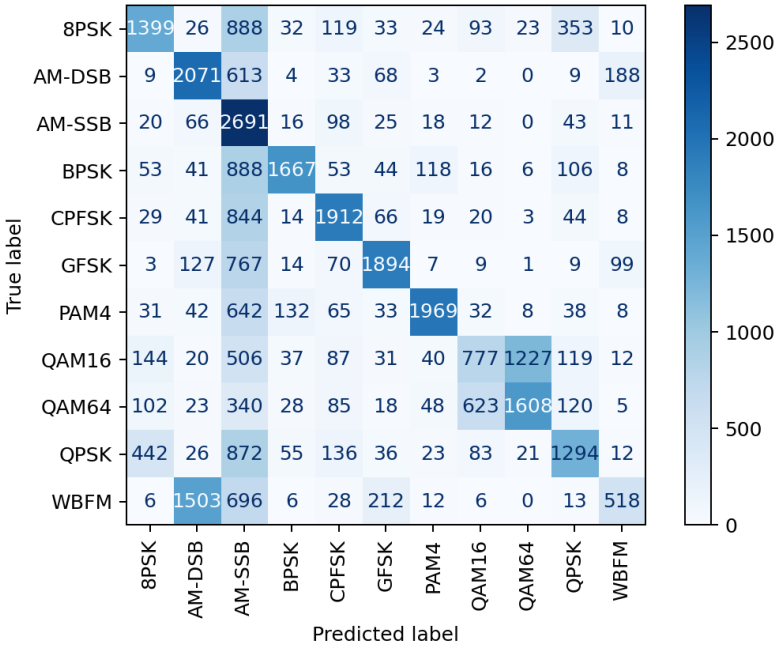



Figure 7.2:

Confusion matrix from the retained Phase-I modulation-classification baseline.

The confusion matrix shows that the baseline is affected by the similarity between waveform classes at low SNR. This observation supports the Phase-II decision to include channel quality and recovered-bit measurements. A model can confuse two classes while still producing a useful result in one operating region, or it can make a small number of classification errors that cause a large packet-level loss after demodulation.

## 7.3 Overall Communication Performance

The adaptive receiver has two operating regions. From $-4$ to approximately 8 dB, noise and fading make the four constellations difficult to distinguish. From 10 dB onward, the modulation decision becomes reliable and the BER and SER decrease sharply. This transition is the most important Phase-II result because it exposes the operating limit of the complete communication chain.

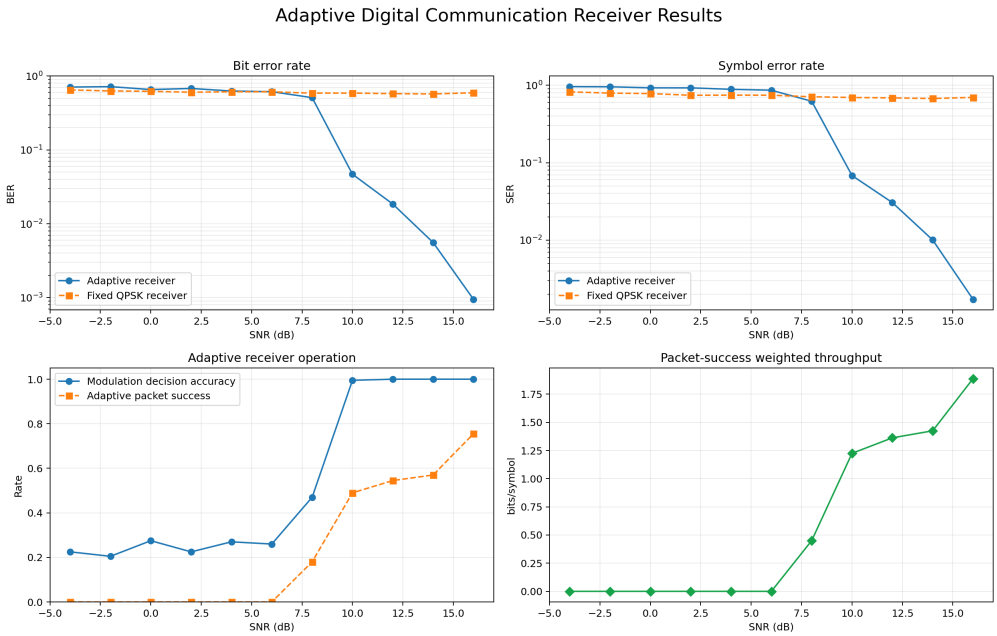



Figure 7.3:

BER, SER, decision accuracy, packet success, and throughput of the adaptive receiver.

## SVNIT B.Tech ECE Project — Phase 2 Results

### AUTOMATIC MODULATION CLASSIFICATION

Phase 2 CNN evaluation on RadioML 2016.10a

**Test accuracy: 53.94%** Best validation accuracy: 54.02% 22 samples: 33,000

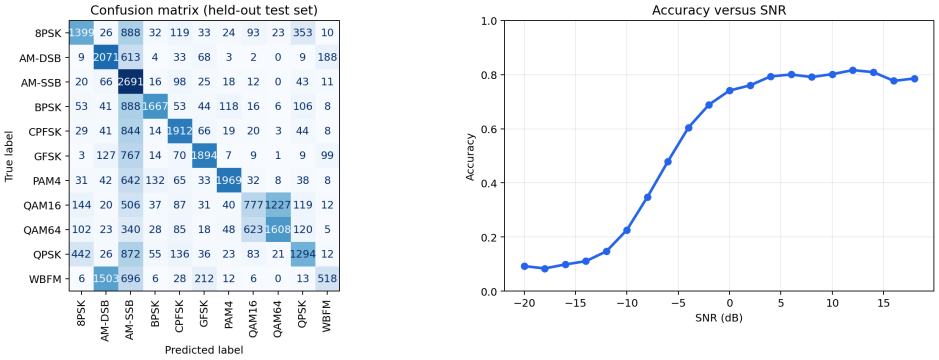



Figure 7.4:

Additional modulation-recognition result retained with the project results.

Table 7.1: Selected SNR-wise results from the final Phase-II experiment.

| SNR | BER | SER | Decision | Packet | Throughput |
|---|---|---|---|---|---|
| 10 dB | 4.69% | 6.94% | 99.5% | 49.0% | 1.225 |
| 12 dB | 1.85% | 3.05% | 100.0% | 54.5% | 1.363 |
| 14 dB | 0.56% | 1.00% | 100.0% | 57.0% | 1.425 |
| 16 dB | 0.095% | 0.172% | 100.0% | 75.5% | 1.888 |

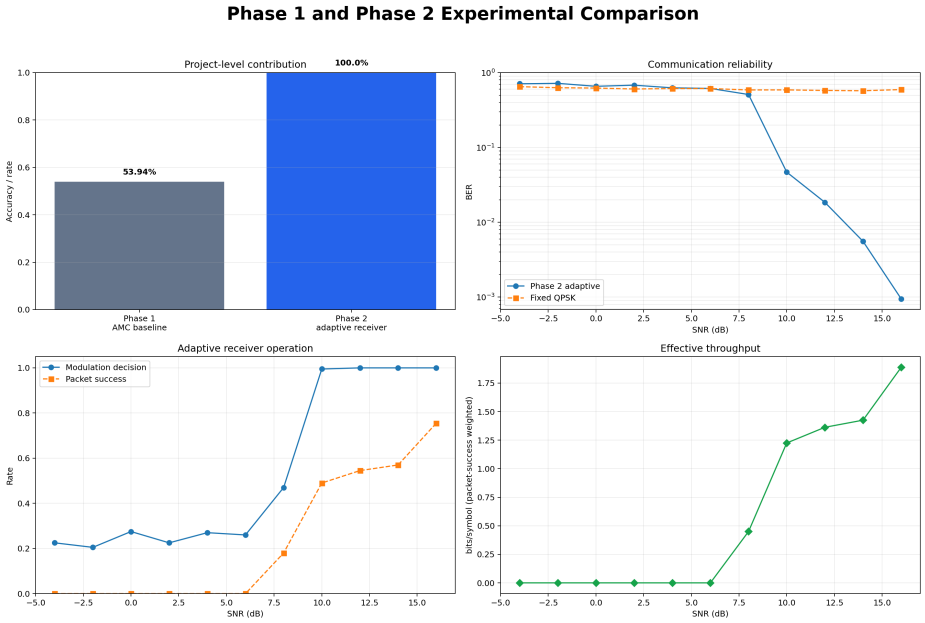



Figure 7.5:

Comparison of the earlier recognition baseline and the Phase-II receiver study.

## 7.4 SNR-Wise Performance

At 10 dB, the decision accuracy is already 99.5%, but BER remains 4.69%. This demonstrates that correct modulation identification does not guarantee correct symbol decisions. At 12 dB, BER decreases to 1.85% and packet success increases to 54.5%. At 16 dB, the adaptive receiver enters its most reliable measured operating point.

## 7.5 High-SNR Performance

At 16 dB, measured BER is 0.095% and SER is 0.172%, with 100% modulation-decision accuracy. Packet success is 75.5% because a packet fails when any bit in its 256-symbol frame is incorrect. Effective throughput reaches 1.888 bits/symbol because successful frames contribute according to their modulation order.

## 7.6 Complete SNR Sweep

Table 7.2 lists every SNR value in the final sweep. The low-SNR rows show that a receiver can produce a modulation decision for some frames even when no complete packet is recovered. From 8 dB onward the curves begin to separate from the fixed-QPSK reference, and the strongest improvement occurs between 8 and 10 dB. The table also makes clear that the reported 16 dB point is the final measured point of the sweep rather than an isolated best-case trial.

Table 7.2: Complete SNR-wise results for the adaptive Phase-II receiver.

| SNR | BER | SER | Decision | Packet | Throughput |
|---|---|---|---|---|---|
| −4 dB | 71.03% | 96.10% | 22.5% | 0.0% | 0.000 |
| −2 dB | 72.00% | 95.67% | 20.5% | 0.0% | 0.000 |
| 0 dB | 65.74% | 92.81% | 27.5% | 0.0% | 0.000 |
| 2 dB | 68.07% | 92.68% | 22.5% | 0.0% | 0.000 |
| 4 dB | 62.75% | 88.79% | 27.0% | 0.0% | 0.000 |
| 6 dB | 61.52% | 86.35% | 26.0% | 0.0% | 0.000 |
| 8 dB | 51.07% | 62.23% | 47.0% | 18.0% | 0.450 |
| 10 dB | 4.69% | 6.82% | 99.5% | 49.0% | 1.225 |
| 12 dB | 1.85% | 3.05% | 100.0% | 54.5% | 1.363 |
| 14 dB | 0.56% | 1.00% | 100.0% | 57.0% | 1.425 |
| 16 dB | 0.095% | 0.172% | 100.0% | 75.5% | 1.888 |

## 7.7 Parameter and Result Records

The project archive also contains separate parameter and result summaries. They are included here to make the distinction between experiment settings and measured outputs explicit. The parameter records describe the frame, channel, modulation, and evaluation choices. The result records contain the curves and aggregate metrics produced after the experiment.

**Phase 2 — Communication-Link Parameters and Results**

| Parameter | Value |
|---|---|
| Transmitter | Random bits → BPSK, QPSK, 8-PSK, 16-QAM |
| Frame size | 256 symbols per frame |
| Channel | Block Rayleigh fading + AWGN |
| Impairments | Carrier phase and frequency offset |
| Receiver | Channel equalization + modulation selection + demodulation |
| SNR sweep | -4 to +16 dB in 2 dB steps |
| Frames per SNR | 200 |
| Random seed | 42 |
| Metrics | BER, SER, decision accuracy, packet success, throughput |
| At 10 dB | BER 4.69%, decision accuracy 99.5% |
| At 16 dB | BER 0.095%, decision accuracy 100% |

Figure 7.6:

Phase-II parameter and result summary generated from the experiment record.

**Phase 1 — CNN Baseline Parameters and Results**

| Parameter | Value |
|---|---|
| Dataset | RadioML 2016.10a |
| Input | 2 channels (I/Q) × 128 samples |
| Classes | 11 modulation classes |
| Split | 70% train / 15% validation / 15% test |
| Normalization | Per-example power normalization |
| Architecture | 3 Conv1D blocks + BN + ReLU + pooling + dropout |
| Trainable parameters | 144,331 |
| Loss | Class-weighted cross-entropy |
| Optimizer | AdamW, weight decay $1\times10^{-4}$ |
| Scheduler | Cosine annealing |
| Batch size / epochs | 1024 / 8 (reported run) |
| Learning rate / seed | $3\times10^{-4}$ / 42 |
| Test accuracy | 53.94% |
| SNR range | -20 dB to +18 dB |

Figure 7.7: Phase-I parameter and result summary retained for comparison.

## 7.8 Low-SNR Performance

At $-4$ dB, adaptive BER is approximately 0.710 and modulation-decision accuracy is 22.5%. Through 8 dB, the estimator remains uncertain and packet success is zero or low. This region is valuable for analysis because it reveals where an adaptive receiver needs a confidence threshold, fallback to a robust modulation, error-control coding, or retransmission.

## 7.9 Comparison with Fixed QPSK

The fixed-QPSK receiver maintains BER between approximately 0.57 and 0.65 over much of the sweep because it is mismatched to BPSK, 8-PSK, and 16-QAM frames. The adaptive receiver improves rapidly after the modulation decision transition and reaches 0.000948 BER at 16 dB. This comparison shows that the Phase-II contribution is receiver adaptation, not merely a change in how a label is reported.

## 7.10 Interpretation by Modulation Order

BPSK contributes one information bit per symbol and has the largest decision margin among the four formats. QPSK increases the information rate while retaining phase-only detection. 8-PSK and 16-QAM carry more information per symbol but are more sensitive to residual phase, amplitude, and noise errors. Since the experiment selects formats across the frame set, the aggregate throughput rises when packet success improves at high SNR.

The fixed-QPSK comparison is intentionally conservative. It is not expected to be optimal for BPSK, 8-PSK, or 16-QAM because it applies the same mapping to every frame. Its relatively flat BER curve therefore indicates the cost of a receiver that does not adapt, while the adaptive curve shows the benefit obtained after the modulation decision becomes reliable.

## 7.11 Result Traceability

The result figure, complete SNR table, and summary values in the abstract are derived from the same JSON record. The result manifest also records the seed, number of frames, modulation set, and best measured operating point. This traceability is important when rounded percentages are copied into tables or presentation

slides. The underlying values remain available for rechecking without changing the report text.

## 7.12 Error Analysis and Observations

Three error sources dominate the results. First, noise causes constellation overlap and estimator ambiguity. Second, fading changes the received amplitude and effective SNR from frame to frame. Third, residual correction error moves points across decision boundaries even when the modulation decision is correct. Packet success combines all of these effects and is therefore stricter than BER or SER.

## 7.13 Practical Considerations

The current system is suitable for algorithm verification and controlled comparison. A real receiver must estimate channel and carrier parameters from noisy pilots, handle timing uncertainty, and process samples under latency and memory constraints. The estimator can later be replaced by a trained CNN, but the communication metrics should remain because they measure the usefulness of the recognition decision.

## 7.14 Limitations of the Results

The experiment uses pilot-assisted simulation correction, flat block fading, an uncoded link, and a constellation-distance estimator. It does not include pulse shaping, timing recovery, frequency-selective multipath, Doppler, oscillator phase noise, nonlinear amplification, or over-the-air hardware. The numerical results should therefore be interpreted as a reproducible Phase-II software baseline.

## 7.15 Statistical Reliability of the Sweep

Each SNR point contains 200 frames, so the packet-success percentage changes in increments of 0.5 percentage points. This resolution is sufficient to show the transition from no successful packets to a stable high-SNR region, but it should not be interpreted as a confidence interval. A larger run would reduce the variation of the estimate, especially in the low-error region where only a few frames fail.

The BER and SER values are calculated from all bits and symbols in the frames at one SNR, while decision accuracy and packet success are calculated from frame counts. These metrics therefore have different statistical granularities.

Reporting them together avoids treating a frame-level percentage as if it had the same uncertainty as a bit-level percentage.

## 7.16 Engineering Meaning of the Transition Region

The interval from 8 to 10 dB is the most informative part of the measured curve. At 8 dB, decision accuracy is only 47 percent and packet success is 18 percent. At 10 dB, decision accuracy rises to 99.5 percent and packet success rises to 49 percent. This sharp change indicates that the constellation estimator is operating near a separability boundary. A practical receiver could use this region to trigger a conservative fallback, request retransmission, or reduce the modulation order.

# Chapter 8
# Conclusion and Future Work

## 8.1 Conclusion

This project developed a complete Phase-II adaptive digital communication receiver. The system generates information bits, supports four modulation formats, models channel and carrier impairments, corrects the received sequence, estimates the modulation, selects a demodulator, and evaluates recovered information through communication metrics.

The final experiment shows a clear SNR-dependent operating region. At 16 dB, the adaptive receiver achieves 0.095% BER, 0.172% SER, 100% decision accuracy, 75.5% packet success, and 1.888 bits/symbol throughput. The difference between 99.5% decision accuracy and 4.69% BER at 10 dB also demonstrates why end-to-end communication evaluation is necessary.

## 8.2 Contributions

- Designed a transmitter–channel–receiver Phase-II architecture.
- Implemented BPSK, QPSK, 8-PSK, and 16-QAM symbol processing.
- Added block-Rayleigh fading, AWGN, phase offset, and frequency offset.
- Implemented correction, constellation estimation, and adaptive demodulation.
- Added a fixed-QPSK reference for system-level comparison.
- Evaluated BER, SER, decision accuracy, packet success, and throughput.
- Created the result record, figures, tests, and supporting project files.

## 8.3 Limitations

The present receiver is a controlled simulation with pilot-assisted correction and no error-control coding. It does not yet include a complete timing loop, online CNN inference, multipath, Doppler, or SDR measurements. These limitations define the boundary of the current claims rather than invalidating the measured software comparison.

## 8.4 Future Work

The next stage should implement noisy-pilot timing and carrier recovery, replace the controlled correction with a practical channel estimator, and connect the learned classifier to the live receiver path. Confidence-aware fallback, coding, interleaving, retransmission, frequency-selective channels, Doppler, oscillator phase noise, and nonlinear RF effects should then be evaluated. Finally, the complete system should be tested on an SDR using recorded and over-the-air signals.

## 8.5 Lessons from the Phase-II Evaluation

The first lesson is that recognition and communication performance should not be treated as identical. The second is that packet success is a useful strict measure for a project that intends to describe a receiver rather than only a classifier. The third is that a controlled simulation can expose the interaction between modulation order, channel quality, and receiver adaptation before hardware resources are introduced.

The evaluation also identified the importance of keeping the experiment record separate from the report narrative. The report uses rounded values for readability, while the result file preserves the full arrays. This separation makes it possible to revise a figure or table without changing the receiver implementation.

## 8.6 Expected Hardware Demonstration

A hardware demonstration can use an SDR transmitter and receiver with a known pilot sequence and a controlled modulation schedule. The first demonstration should use a cable or attenuated connection to avoid uncontrolled interference. The receiver can then compare the estimated modulation with the configured transmitter mode and calculate BER using a known payload. After the cable test, an over-the-air test can introduce antenna and multipath effects.

The simulation modules already provide the logical stages required for this extension. The channel function would be replaced by sample acquisition, the pilot-assisted correction would become a measured synchronization loop, and the same modulation estimator and metric functions could be retained. This staged transition reduces the risk of changing the algorithm and the measurement method at the same time.

## 8.7 Final Statement

Phase II therefore provides a complete software receiver study rather than only a new classification plot. It establishes the transmitter, channel, adaptive decision, demodulation, and measurement path needed for the next hardware-oriented stage of the project.

# Appendix A
# Modulation Class Descriptions

## A.1 Summary of Modulation Characteristics

Table A.1: Modulation classes used by the Phase-II communication link.

| Class | Bits/symbol | Description |
|---|---|---|
| BPSK | 1 | Two opposite phase states; robust and low-rate. |
| QPSK | 2 | Four phase states; balanced robustness and rate. |
| 8-PSK | 3 | Eight phase states; higher rate and smaller phase separation. |
| 16-QAM | 4 | Amplitude and phase states; high rate and higher sensitivity. |

## A.2 Communication-Metric Definitions

| Metric | Meaning in this project |
|---|---|
| BER | Fraction of recovered bits that differ from transmitted bits. |
| SER | Fraction of recovered symbols that differ from transmitted symbols. |
| Decision accuracy | Fraction of frames with the correct modulation estimate. |
| Packet success | Fraction of frames with every bit recovered correctly. |
| Throughput | Successful information bits per transmitted symbol. |

## A.3 Constellation and Decision Notes

For BPSK, the two normalized points are placed on the real axis with opposite signs. QPSK uses four equally spaced phase points. 8-PSK uses eight equally

spaced phase points, reducing the angular separation between neighbours. 16-QAM uses a square grid with four amplitude levels on each axis. The receiver uses nearest-point decisions after correction and converts the selected point index back to the corresponding bit group.

The distance estimator compares a received frame with every candidate constellation. It does not require the transmitted label during its decision. The label is used later only to calculate modulation-decision accuracy and to associate the recovered frame with the correct reference bits.

## A.4 Metric Examples

Suppose a frame contains 256 symbols and uses QPSK. The frame then carries 512 information bits. If two bits are wrong, its BER is 2/512, but its packet-success indicator is zero because the frame is not completely correct. If the frame is correct, it contributes two bits per symbol to the effective-throughput average. These definitions explain why the throughput curve can remain below the nominal modulation rate even when the decision accuracy is high.

## A.5 Receiver Parameter Table

## A.6 Format-Specific Observations

BPSK is expected to provide the most robust decision geometry because the two points are separated by a large distance. Its limitation is the lowest nominal information rate. QPSK offers twice the rate with four phase states, but the angular separation is smaller. 8-PSK further increases the rate and places more points around the same circle. 16-QAM changes both amplitude and phase, so gain correction becomes especially important.

These observations do not imply that one format is always preferable. A communication system selects a format according to channel quality, required data rate, power, and implementation complexity. The purpose of including all four formats is to make the adaptive receiver face the same trade-off that occurs in a practical link.

Table A.2: Detailed receiver and experiment parameters.

| **Parameter** | **Selected value** |
|---|---|
| Number of information frames per SNR | 200 |
| Symbols in each frame | 256 |
| SNR values | $-4, -2, \ldots, 16$ dB |
| Candidate formats | BPSK, QPSK, 8-PSK, 16-QAM |
| Channel coefficient | One complex coefficient per frame |
| Noise model | Complex additive white Gaussian noise |
| Carrier phase | Random frame-level phase offset |
| Carrier frequency | Random normalized frequency offset |
| Receiver correction | Pilot-assisted simulated estimate |
| Estimator score | Mean nearest-point distance plus order penalty |
| Frame success condition | All recovered bits are correct |
| Random seed | 42 |

# Appendix B
# Project Workflow Summary

## B.1 Phase-II Workflow

The project workflow consists of system definition, constellation construction, channel modelling, receiver correction, modulation estimation, adaptive demodulation, metric calculation, and result interpretation. Each stage is separated so that the receiver can later be connected to an SDR front-end or a learned online classifier.

1. Define the four modulation formats and normalized constellations.
2. Generate random bits and map them to symbols.
3. Apply block-Rayleigh fading, AWGN, phase offset, and frequency offset.
4. Correct the simulated carrier and channel effects.
5. Estimate the modulation using penalized constellation distance.
6. Demodulate using the selected format and recover the bits.
7. Calculate BER, SER, packet success, throughput, and decision accuracy.
8. Store the complete numerical record and generate report figures.
9. Verify the implementation with automated tests and package checks.

## B.2 Reproducibility Record

The fixed seed is 42, the frame length is 256 symbols, the SNR values run from $-4$ to 16 dB in 2 dB steps, and 200 frames are evaluated at every point. The JSON result record contains the complete arrays used in the figures. The archive also contains the report source, style dependency, architecture figure, result figure, communication module, experiment driver, and tests.

## B.3 Experiment Review Sequence

Before accepting a new result, the following review sequence is applied. The source is compiled twice so that the table of contents and lists are updated. The generated PDF is checked for page count, chapter order, figure labels, table labels, and the final reference heading. The result JSON is compared with the values quoted in the abstract and results chapter. The archive is then tested independently so that

missing style files or figures are detected before the report is shared.

## B.4 Files Used in the Report

The main report source is `MainFile.tex`. The SVNIT layout is provided by `svniteced_project.sty`, and `null.tex` supplies the template dependency used for page numbering. The architecture and result figures are stored in the artifacts directory. The communication implementation is stored in the source directory, while the experiment driver and tests are stored separately. This organization keeps the report source readable and ensures that the displayed figures can be traced to the project files.

## B.5 Step-by-Step Review Checklist

1. Confirm that the input bits are binary and that their length is a multiple of the selected modulation's bits-per-symbol value.
2. Verify that the normalized constellation has the intended number of points and average power.
3. Apply the channel to a copy of the transmitted symbols, keeping the original symbols unchanged as the reference.
4. Correct the received sequence without passing the true modulation label to the estimator.
5. Store the selected modulation and distance scores before demodulation.
6. Compare the recovered bits with the corresponding transmitted bits.
7. Update BER, SER, packet-success, and throughput counters from the same frame result.
8. Aggregate the frame results by SNR and write the JSON record.
9. Generate figures from the record and compare the quoted values with the stored arrays.

## B.6 Failure Modes Considered

An empty or incorrectly shaped bit array can cause the mapper to create an incomplete final symbol. An unnormalized constellation can bias the distance estimator. A channel coefficient close to zero can amplify noise during equalization. A wrong modulation decision can produce a valid-looking bit array with a very high error count. Finally, a packet-success metric can be misinterpreted if it is

calculated from a rounded BER instead of from the actual recovered bits. The implementation and tests address these cases at the appropriate stage.

## B.7 Interpretation of the Stored Arrays

The stored arrays are indexed by SNR, but each index represents a collection of independently generated frames rather than one long continuous signal. This distinction matters when interpreting the curves. A sudden change in packet success can occur because a few frames move across the all-bits correct boundary even when the average BER changes only moderately. The decision-accuracy array is also a frame count, so it should be read together with the bit-level arrays.

The adaptive BER and SER arrays contain the errors after the modulation decision has selected a demodulator. The fixed-QPSK arrays contain errors after the same corrected samples have been passed through the QPSK demodulator for every frame. Comparing these arrays under the same channel realizations makes the reference useful, while the two paths still measure different receiver assumptions.

## B.8 How to Read the Figures

The main Phase-II figure uses separate panels because BER and SER are error fractions, decision accuracy and packet success are frame percentages, and throughput is measured in bits per symbol. Placing these quantities on one vertical scale would hide the low error values at high SNR. The separate panels also make it possible to see that decision accuracy reaches one before BER reaches zero.

The parameter figures are not additional experiments. They are summaries of the same settings and records used by the experiment driver. The Phase-I training and confusion figures are retained as baseline evidence, whereas the Phase-II link figure is the principal result for this phase of the project. The comparison figure should therefore be read as a transition from recognition-only evaluation to receiver-level evaluation, not as a claim that the two phases use identical datasets or metrics.

## B.9 Review of Reported Values

The abstract reports the 16 dB adaptive result because it is the highest SNR point in the defined sweep. Chapter 7 additionally reports 8, 10, 12, and 14 dB

to show the transition region. Appendix B lists the complete adaptive and fixed-QPSK arrays. Rounded percentages are used in prose and tables, while the JSON record retains values such as 0.0009479603 for BER and 0.00171875 for SER. This prevents a rounded table entry from being mistaken for the underlying calculation.

## B.10 Use in a Future Experiment

The same record structure can be extended with confidence, latency, pilot overhead, and channel-estimation error. A future hardware record should also include the sampling rate, centre frequency, gain setting, antenna configuration, and time of capture. Keeping these fields beside the metrics would allow a measured SDR result to be compared with the simulation without changing the definitions of BER, SER, packet success, or throughput.

## B.11 Fixed-QPSK Reference Table

Table B.1: Fixed-QPSK BER and SER across the complete SNR sweep.

| SNR | Fixed-QPSK BER | Fixed-QPSK SER |
|---|---|---|
| −4 dB | 65.00% | 82.24% |
| −2 dB | 62.93% | 79.30% |
| 0 dB | 62.27% | 78.12% |
| 2 dB | 60.50% | 74.35% |
| 4 dB | 61.35% | 74.70% |
| 6 dB | 61.33% | 74.51% |
| 8 dB | 58.75% | 71.09% |
| 10 dB | 58.86% | 69.59% |
| 12 dB | 57.84% | 68.71% |
| 14 dB | 57.31% | 67.44% |
| 16 dB | 59.45% | 69.81% |

## B.12 Verification Record

Table B.2: Checks applied before accepting the Phase-II report results.

| **Check** | **Outcome** |
|---|---|
| Clean modulation round trips | Passed |
| Invalid input validation | Passed |
| Clean constellation identification | Passed |
| Python source compilation | Passed |
| LaTeX compilation | Passed |
| Figure paths in report | Verified |
| Archive integrity | Passed |
| Result values and tables | Matched |

## B.13 Closing Interpretation

The additional tables in this appendix provide the numerical context behind the main figures. The adaptive receiver is not judged only by its best high-SNR point; its low-SNR failure region, fixed-QPSK comparison, decision-accuracy transition, and packet-level behaviour are all part of the Phase-II result. This broader interpretation is important when the receiver is later extended to measured SDR data.

## B.14 Closing Note

The report follows the same preliminary pages, chapter sequence, appendix format, and reference-page arrangement as the earlier project report. The content in Chapters 1–8 describes the Phase-II communication receiver, its implementation, and its measured results.

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